

Fig. 1.15 Typical methods for modelling a fixed length domain. *Left*: particles are introduced at fixed locations $x = 0$ and $x = L$; *right*: in periodic boundary conditions, potentials are introduced to model the effect of an atom just outside the domain (on either end)

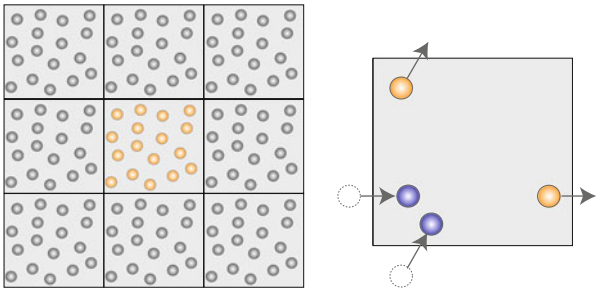


Fig. 1.16 Periodic boundary conditions illustrated in two dimensions. *Left*: the atom locations of the central simulation cell are repeated in each neighboring cell; potential energy contributions must be included not only for atom-atom interactions within the simulation cell but also for interactions between atoms of the central cell and those in the neighbors. *Right*: as an atom leaves the central region, it reenters via the opposite boundary

If a particle moves to the right of $x = L$ we simply shift its position to $x-L$; likewise any particle exiting to the left of $x = 0$ has its position shifted to $x+L$ (see Fig. 1.15). This type of boundary condition is a common choice and considered more relevant in simulations of realistic systems than confining boundary conditions. Periodic boundary conditions allow us to preserve Newton's third law, the translational symmetry, and thus the conservation of momentum.

When dealing with crystalline materials, or for initializing more general systems, the atoms are typically confined for most of their evolution to small neighborhoods of a collection of periodic points called a lattice. On the line, we think of a (finite) lattice as a sequence of discrete points separated by a fixed distance Δx. An obvious benefit of using periodic boundary conditions is that, with a uniform pair potential, the energy minimizers are points of a regular lattice; with confining potentials this is unlikely to be the case.

The technique of periodic boundary conditions can be extended to 2D or 3D (Fig. 1.16). We assume our simulation box is surrounded not by a large number of particles moving on completely independent paths, but, rather, that our box is bordered on each face by a box of particles whose positions are exact shifted images of those of the resolved system. Atoms moving out of the central box are not reflected back into the central domain, but enter the box at a corresponding point on the opposite face. For purposes of illustration we have diagrammed the situation for a planar region. If φ_{ij} is the potential energy function for interactions between